12) Mini batch Gradient Descent

```
import numpy as np
X = np.array([1,2,3,4,5])
Y = np.array([2,4,6,8,10])
W=0
b=0
lr =0.01
epochs = 50
batch_size =2
for epoch in range(epochs):
    for i in range(0,len(X), batch_size):
        X_batch =X[i: i+batch_size]
        Y_batch = Y[i: i+batch_size]
        Y_pred = W* X_batch + b
        dW = -2* np.mean(X_batch * (Y_batch - Y_pred))
        db =-2* np.mean(Y_batch - Y_pred)
        W= W-Lr*dW
        b = b-Lr*db
print("Weight:",W)
print("Bias:", b)
```

OUTPUT

Weight : 1.9310260387911284

Bias: 0.3030957122303656